

# The idea of the integers

The naturals, their opposites and zero — one set closed under subtraction.

LESSON 17–18

2 × 40 MIN

MATEMATIKA 7, §6

S8 1.1

LESSON CLOCK

15'

25'

25'

20'

5'

|                            |        |
|----------------------------|--------|
| The three sets             | 15 min |
| Closure                    | 25 min |
| Factors, multiples, primes | 25 min |
| Recognising integers       | 20 min |
| Homework                   | 5 min  |

LEARNING OBJECTIVES

- Define the set of integers and use the symbol  $\mathbb{Z}$ .
- Distinguish natural numbers, whole numbers and integers.
- Recall factors, multiples and primes among the naturals.
- Decide whether a given number is an integer.

TEXTBOOK

|           |                  |
|-----------|------------------|
| National  | pp. 42–46        |
| Cambridge | Stage 8, pp. 2–6 |
| Workbook  | Exercise 1.1     |

01

## Explanation

### The three sets

| SET             | SYMBOL       | MEMBERS                   |
|-----------------|--------------|---------------------------|
| natural numbers | $\mathbb{N}$ | 1, 2, 3, ...              |
| whole numbers   | —            | 0, 1, 2, 3, ...           |
| integers        | $\mathbb{Z}$ | ..., -2, -1, 0, 1, 2, ... |

Every natural number is an integer, and every integer is a rational number (the next chapter). Written with the subset symbol:  $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q}$ .

$\mathbb{Z}$  COMES FROM THE GERMAN *ZAHLEN*

Meaning “numbers”.  $\mathbb{N}$  is for *naturalis* and  $\mathbb{Q}$  for *quoziante*. The letters are international, and they will be used for the rest of school mathematics.

### Closure

A set is **closed** under an operation if the answer always stays inside the set.

| OPERATION      | $\mathbb{N}$      | $\mathbb{Z}$ |
|----------------|-------------------|--------------|
| addition       | closed            | closed       |
| multiplication | closed            | closed       |
| subtraction    | <b>not</b> closed | closed       |
| division       | not closed        | not closed   |

The third row is the whole reason for the chapter:  $3 - 5$  leaves  $\mathbb{N}$  but stays inside  $\mathbb{Z}$ .  
The fourth row is why  $\mathbb{Q}$  will be needed.

6 ÷ 3 = 2 DOES NOT MAKE  $\mathbb{Z}$  CLOSED UNDER DIVISION

Closure requires **every** case to work. One counter-example —  $3 \div 2$  — settles it, just as one counter-example disproved a claim in the algebra course.

Factors, multiples, primes

| WORD            | MEANS                    | EXAMPLE            |
|-----------------|--------------------------|--------------------|
| factor of $n$   | divides $n$ exactly      | 1, 2, 3, 4, 6, 12  |
| multiple of $n$ | $n$ times a whole number | 12, 24, 36, ...    |
| prime           | exactly two factors      | 2, 3, 5, 7, 11, 13 |
| composite       | more than two factors    | 4, 6, 8, 9, 10     |

1 is neither prime nor composite: it has only one factor. 2 is the only even prime.

EVERY WHOLE NUMBER GREATER THAN 1 IS A PRODUCT OF PRIMES, IN ONE WAY ONLY

$60 = 2^2 \times 3 \times 5$ , and no other set of primes gives 60. This is the fundamental theorem of arithmetic, and it underlies HCF, LCM and every fraction calculation.

Recognising integers

| NUMBER        | AN INTEGER? | WHY                     |
|---------------|-------------|-------------------------|
| $-7$          | yes         | a negative whole number |
| $0$           | yes         | —                       |
| $\frac{6}{3}$ | yes         | it equals 2             |
| $\frac{7}{2}$ | no          | 3.5                     |
| $-4.0$        | yes         | it equals $-4$          |
| $0.333\dots$  | no          | $\frac{1}{3}$           |

LOOK AT THE VALUE, NOT AT THE WAY IT IS WRITTEN

$\frac{6}{3}$  is written as a fraction but **is** the integer 2. A number is an integer if it equals one, however it happens to be written.

## 02 Worked examples

**EXAMPLE 1** Which of  $-7$ ,  $0$ ,  $\frac{6}{3}$ ,  $\frac{7}{2}$ ,  $-4.0$ ,  $0.333\dots$  are integers?

①  $-7$  and  $0$  plainly are.

②  $\frac{6}{3} = 2$  — an integer.

Look at the value.

③  $\frac{7}{2} = 3.5$  — not.

④  $-4.0 = -4$  — yes;  $0.333\dots = \frac{1}{3}$  — no.

ANSWER

$-7$ ,  $0$ ,  $\frac{6}{3}$ ,  $-4.0$

**EXAMPLE 2** List all the factors of 36, and say whether it is prime.

① Pair them:  $1 \times 36$ ,  $2 \times 18$ ,  $3 \times 12$ ,  $4 \times 9$ ,  $6 \times 6$ .

②  $1, 2, 3, 4, 6, 9, 12, 18, 36$

Nine factors.

③ More than two factors.

④ Composite.

ANSWER

Nine factors; composite

**EXAMPLE 3** Is  $\mathbb{Z}$  closed under subtraction? Under division?

- 1 Subtraction:  $a - b$  is always an integer.  
Closed.
- 2 Division:  $3 \div 2 = 1.5$ .
- 3 Not an integer.
- 4 Closed under subtraction, not under division.

**ANSWER**

Yes; no

**03** Interactive model

Ask the class to find a subtraction of naturals with no natural answer, then to find one of integers with no integer answer; the second search fails, and closure is understood.

**Which set does it belong to?**

7 is:

a natural number and an integer

an integer only

neither

negative

$-7$  is:

a natural number

an integer but not natural

neither

a fraction

0 is:

an integer

not a number

negative

natural in every convention

$-3.5$  is:

an integer

not an integer

natural

positive

$+5$  and  $5$  are:

different numbers

the same number

opposites

both negative

The opposite of  $-12$  is:

$12$

$-12$

$0$

$\frac{1}{12}$

How many integers lie between  $-2$  and  $2$ ?

2

3

4

infinitely many

A temperature of  $-8^{\circ}$  is written with a minus because:

it is small

it is below zero

it is an error

it is a fraction

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.



| ENGLISH          | O'ZBEKCHA    | РУССКИЙ                |
|------------------|--------------|------------------------|
| Integer          | Butun son    | Целое число            |
| Natural number   | Natural son  | Натуральное число      |
| Set              | To'plam      | Множество              |
| Factor           | Bo'luvchi    | Делитель               |
| Multiple         | Karrali      | Кратное                |
| Prime number     | Tub son      | Простое число          |
| Composite number | Murakkab son | Составное число        |
| Closed under     | Yopiq        | Замкнутое относительно |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

$\mathbb{Z}$  denotes:

the naturals

the integers

the rationals

the reals

Is 0 a natural number in this course?

yes

no

sometimes

undefined

$\mathbb{Z}$  is closed under:

division

subtraction

both

neither

Is 6 3 an integer?

yes

no

sometimes

undefined

How many factors has a prime?

1

2

3

many

Is 1 prime?

yes

no

sometimes

undefined

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Is  $-7$  an integer?

ANSWER Yes

2. Is  $3.5$  an integer?

ANSWER No

3. Is  $\frac{8}{4}$  an integer?

ANSWER Yes

4. Is  $0$  an integer?

ANSWER Yes

5. Factors of  $12$

ANSWER  $1, 2, 3, 4, 6, 12$

6. The first five primes

ANSWER  $2, 3, 5, 7, 11$

7. Is  $1$  prime?

ANSWER No

Medium · the standard question

7 PROBLEMS

## 1. Factors of 36

ANSWER 1,2,3,4,6,9,12,18,36

## 2. Is 36 prime or composite?

ANSWER Composite

## 3. Prime factorisation of 60

ANSWER  $2^2 \times 3 \times 5$ 

## 4. Prime factorisation of 84

ANSWER  $2^2 \times 3 \times 7$ 5. Is  $\mathbb{N}$  closed under subtraction?

ANSWER No

6. Is  $\mathbb{Z}$  closed under subtraction?

ANSWER Yes

7. Is  $\mathbb{Z}$  closed under division?

ANSWER No

## Hard · stretch and reason

7 PROBLEMS

1. HCF of 36 and 60

ANSWER 12

2. LCM of 36 and 60

ANSWER 180

3. How many factors has 60?

ANSWER 12

4. How many factors has  $2^4 \times 3^2$ ?

ANSWER 15

5. The only even prime

ANSWER 2

6. Every integer is rational: true?

ANSWER Yes —  $n = \frac{n}{1}$

7. Is the set of odd numbers closed under addition?

ANSWER No — the sum is even

## 07 Homework

### Homework — 5 tasks

Justify every “no” with a counter-example.

- Which of  $-3$ ,  $2.5$ ,  $\frac{9}{3}$ ,  $0$ ,  $-1.5$  are integers?
- List all the factors of 48 and say whether it is prime.
- Write the prime factorisation of 90.
- Find the HCF and the LCM of 24 and 36.

5. Is the set of even numbers closed under addition? Under division?